AI

Everyone Is Talking About AI. No One Tells You What You Need to Learn

The AI skills to learn, in the right order, for data and tech careers. What to learn, what to skip, and a 90-day plan.

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Pick your level first

People stall because they try to learn everything. Most jobs need level 1 and 2.

- **Level 1: Use AI well (everyone)**

  Clear prompts, check every output, keep company data private, know where AI is weak.

- **Level 2: Work with AI (data and tech roles)**

  Use AI for SQL, Python and analysis. Understand tokens, context window and hallucination.

- **Level 3: Build with AI (builders)**

  Call models from code, get JSON output, embeddings, RAG, agents, evaluation.

- **Not needed for most jobs**

  Training large models from scratch. That is research work.

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Level 1: Use AI well

Use these 5 parts for almost any prompt. The last one catches most mistakes.

```
ROLE:    You are a senior data analyst.
CONTEXT: Table sales has order_id,
         order_date, region, amount.
TASK:    Month-on-month revenue growth %
         by region.
FORMAT:  PostgreSQL, one line per step.
CHECK:   List 2 ways this could be wrong.
```

Check every output

AI can be confidently wrong. Verify numbers, names and code.

Protect data

Never paste confidential data into a public AI tool. Follow your company's AI policy.

Your basics matter more now

SQL, Python, statistics and business sense let you judge what AI gives you.

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Level 2: How AI works

The 5 terms that come up at work and in interviews, in plain words.

LLM

Large language model. Predicts the next piece of text based on patterns learned from huge amounts of text.

Token

The unit of text a model reads and writes, roughly part of a word. Usage is counted in tokens.

Context window

How much text the model can consider at once, including your prompt and its reply.

Temperature

Randomness setting. Lower gives more consistent answers, higher gives more varied ones.

Hallucination

A confident answer that is wrong. The reason checking output is a core skill.

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Level 3 and your 90-day plan

Build skills in order, one hour a day.

Days 1-15

Use AI daily at work with the 5-part prompt. Check every output.

Days 16-45

Strengthen SQL, Python and statistics. Use AI as a tutor, not an answer key.

Days 46-60

Learn the 5 terms. Use AI to classify 50 rows of real text, then check accuracy by hand.

Days 61-90

Call a model from Python, get JSON output, and build a small RAG search: find relevant documents first, then let the model answer from them.

Skip for now: training models from scratch, deep transformer maths, chasing every new tool.

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